



Gorilla Glass 4

Would Gorilla Glass 4 still make you skip a heartbeat when you drop your smartphone? We will soon know <http://dgit.in/GorillaG4>



Badass Coffee maker

Do you really need a bomb-proof and rugged coffee maker? Coming from Kickstarter, Coffeeboxx might catch your eye. <http://dgit.in/Coffeeboxx>

Bazaar

AMD Radeon R9 285

Tonga chipset is more of a feature upgrade

We finally got our hands on the R9 285 which was launched back in September by AMD as part of their 30th year celebrations. The card gives off an impression that it might be faster than the R9 280, after all, it has been named the R9 285. But as we take a look at the specifications we wonder if that can be the case. The R9 280 and the R9 285 seem to have very similar specifications but the main differences are that the R9 285 is using GCN 1.2 which is an upgrade over the GCN 1.0 that the R9 280 uses. With improvements in the architecture the GPU is more efficient with memory bandwidth and gets away with less of it compared to the R9 280. Furthermore, the TDP for the R9 285 is 190 W while the R9 280 is rated for 250 W. All of this tells us that there have been multiple trade-offs in favour of greater efficiency overall. But generally, this means performance doesn't get a big boost.

For gamers, there has been a marked improvement in tessellation capabilities of the card. This improvement only shows when the card is stressed otherwise at lower loads higher SKUs perform better. So if subjected to a gradual increment of tessellation load, the card will continue to process more efficiently long after higher SKUs start throttling down. The next big feature improvement is related to video encoding and decoding. With higher resolution gaining mainstream popularity graphics cards have to focus on crunching more pixels efficiently rather than letter the same old



molex connectors which is more than sufficient to handle 190W for a few overclocking sessions. During our benchmarks the temperature peaked at 77 degrees celsius during peak load and while idle it was around 42 degrees celsius. We didn't get to test the reference R9 280 so we can't compare it to that. But given all the third-party vendor cards that we did test, we can safely say that the temperatures were in a similar range. So we can attribute the lower temperatures to the improvement in the architecture rather than the stock cooler.

When we look at the performance figures we see that the card performs similar to a R9 270X, so from amongst the cards that we've tested this would be the ASUS R9 270X DC2T and looking at the cards from NVIDIA the figures were close to that of the MSI GTX 760TF. FPS for Bioshock Infinite was off by 20 when at Ultra configuration but everything else seemed to be well within the margin of error. So the numbers were nowhere close to the R9 280 as one would infer from the model number. OpenGL

performance was better than the R9 270X but OpenCL numbers were lower than the R9 270X by about 35 points which is quite surprising given that the architectural changes should have improved compute performance or at least have kept it at the same level as before.

Mithun Mohandas



mechanics scale with the load. The R9 285 has an improved Unified Video Decoder which has paved the way for better encoding and decoding performance at higher resolutions.

Being a reference card we weren't expecting any groovy cooling just the bare minimum and rightly so, that's what we got. The card uses two 6-pin



Performance..... 54
Value 75
Build 60

Specifications

Chipset: Tonga; **Base clock:** 1050MHz; **Memory clock:** 1800MHz; **Stream processors:** 1792; **Texture Units:** 112; **ROPs:** 32; **Manufacturing process:** 28nm, PCIe 3.0, 4096 x 2160 digital resolution support, 2 GB Memory; **DirectX support:** 12; **OpenGL support:** 4.4; **Power Connectors:** 6Pin + 6Pin; **TDP:** 190W

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